

A convenient fibration for dependently-typed probability theory

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slides:



paper:



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What have we done?

Semantic **model** for **dependent types** $\Gamma, x : A \vdash B$

best case scenario: (split full comprehension category [Jacobs'93])

$\Gamma \vdash (x : A) \rightarrow B$	$\Gamma \vdash (x : A) \times B$	
dependent functions	dependent pairs	
(Π -types)	(Σ -types)	
$\Gamma \vdash M ::= K$	$\Gamma \vdash \{x : A \mid \varphi\}$	$a, b : \mathbb{R} \vdash (a, b]$
propositional equality	subspaces	continuous spaces
(extensional)	(refinement types)	

where **construction** means (quasi-)measurable map

What have we done?

Fibred Giry monad (split, in fact)

[Lawvere'62, Giry'80]

$$\begin{array}{lll} \Gamma \vdash \mathbf{D}A & \Gamma \vdash \mathbf{P}A := \{\mu : \mathbf{D}A \mid \mu(A) = 1\} & (\Gamma \vdash A) \\ \text{distributions / measures} & \text{probability distributions} & \\ \text{(non-normalised)} & & \end{array}$$

supporting continuous probability with the usual properties:

(fibred variant of Kock's ['12] **synthetic measure theory**)

$$\begin{array}{ll} \Gamma, \mu : \mathbf{D}A, f : A \rightarrow [0, \infty], \nu : \mathbf{D}B, g : B \rightarrow [0, \infty] \vdash & \\ a, b : \mathbb{R}, a < b \vdash & \\ \mathbf{U}_{(a,b]} : \mathbf{P}(a,b] & \\ \text{uniform distribution} & \end{array} \quad \begin{array}{l} \int d\mu f \int d\nu g = \iint (\mu \otimes \nu)(dx, dy) (f x) \cdot (f y) \\ = \int d\nu g \int d\mu f \\ \text{(fibred Fubini's theorem)} \end{array}$$

What have we done?

Dependently-typed probability

Case study: probabilistic concept inhabiting inherently dependent space:

$$\Gamma \vdash \mathbb{E} : \begin{array}{c} \text{observation} \quad \mu\text{-integrable r.v.} \\ (\mu : \mathbf{D}\Omega) \rightarrow (H : \Omega \rightarrow \Theta) \rightarrow (f : \mathcal{L}^1(\Omega, \mu)) \rightarrow \\ \{g : \mathcal{L}^1(\Omega, \mu_H) \mid g = \mathbb{E}_\mu[f \mid H]\} \quad \text{(standard } \Theta) \\ \text{conditional expectation} \end{array}$$

by extracting **construction** from standard proof

What have we done?

Best possible situation:

- ▶ Semantic model for dependent types (Π , Σ , extensional identity, subspaces)
(split fibration)
- ▶ Semantic model for dependently-typed probability
(split distribution and probability monads)

Case study

- ▶ dependently-typed probability
(measurability of conditional expectation under regularity assumptions)

Why do it?

Dependent types trim ill-defined edge cases

- **Urn models** indexed by their total number of balls:

[Jacobs'22]

$$n : \mathbb{N} \vdash \mathbf{Urn}_n := \{(b, r) : \mathbb{N} \times \mathbb{N} \mid b + r = n\}$$

- Avoid '**almost-sure** safety', obtain actual '**safety**':

$$a, b : \mathbb{R} \vdash (w \sim \mathbf{U}_{(0,1]}; \frac{1}{w}) : \mathbf{P}[1, \infty)$$

Why do it?

Community interest

- ▶ Stream-sampling for probabilistic programming using continuous functions via **compactly generated Hausdorff spaces** [Dahlqvist, Silva, and Smith'23]
- ▶ **Synthetic probability** for dependent types with Markov fibrations [Rischel'26]
- ▶ Probabilistic **graphical models** with the codomain fibration [St Clere Smithe '24, '25]

Personal interest

- ▶ Dependently-typed **Probabilistic programming** and **Bayesian inference** (WIP)
(use Brady's ['21] Idris 2)
- ▶ Unsatisfactory early version of use-case using **HOL** and **refinement types**
CIRM tutorial [Kammar'22, unpublished]

Why is it tricky?

Aumann's theorem ['61]

No simply-typed measurable-function measurable space, let alone dependent-functions

Locally Cartesian-closed categories

Codomain fibrations of sheaf models for probability model full dependent types
several models for classical continuous probability
quasi-Borel spaces and their precursor [Staton et al. '16+'17]
probability sheaves [Simpson'17+'24]
nominal techniques [Li, Aytac, Johnson-Freyd, Ahmed, Holtzen'24]
still need a fibred probability monad!
(codomain fibration also not **split**—
inconvenient for external development in Staton et al. and Simpson models)

Why is it tricky?

What should $\Gamma \vdash \mathbf{P}A$ be?

Non-answer:

$$\gamma : \Gamma \vdash \mathbf{X}A := \left\{ \mu : \mathbf{P}(\Gamma, x : A) \mid \mu(\mathrm{d}\gamma', x \mapsto a)\text{-a.s. } \gamma' = \gamma \right\}$$

$\mathbf{X}A$ is a monad on each fibre, but reindexing/substitution does not preserve it:

$$(\mathbf{X}A) [\theta] \not\cong \mathbf{X}(A [\theta])$$

$\mathbf{X}A$ isn't **fibred monad**

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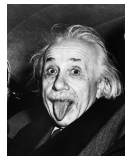
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$\mathbf{X}A$ isn't **fibred monad**

Danel and I listened to **Einstein**:

“If you are out to describe the truth, leave elegance to the
tailor.”

We started developing semantics for dependent types with
non-fibred monads.



Why is it tricky?

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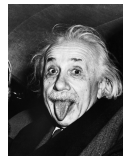
$\mathbf{X}A$ isn't **fibred monad**

Danel and I listened to **Einstein**. We should have listened to **Rasmus**:



“If you are out to describe the truth, leave elegance to the tailor.”

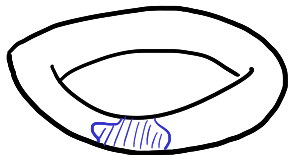
“Danel and Ohad, your pain threshold is much too high for your own good.”



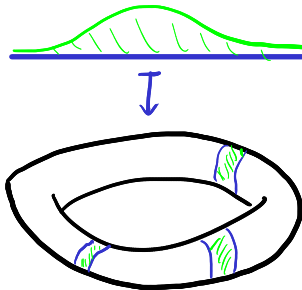
Talk structure

- ▶ Motivation
- ▶ quasi-Borel spaces & families
- ▶ fibred probability monads
- ▶ dependently-typed probability: conditional expectation

measurable spaces



quasi Borel spaces



primitive notions: points & **admissible events**
(measurable subsets)

points & **admissible random elements**

derived notions: **admissible random elements** $\alpha : \mathbb{R} \rightarrow M$

admissible events $E \in \mathcal{B}_X$

Metaphorology¹ $\mathcal{R}$

over set $\underline{X}$ of **points**: subset of functions—**admissible random elements**— $\alpha : \mathbb{R} \rightarrow \underline{X}$ closed under:

constants:

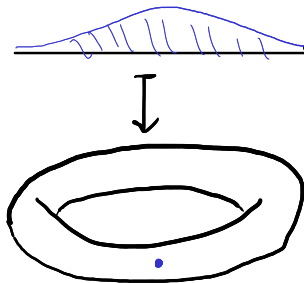
$$\frac{x \in \underline{X}}{\underline{x} := (\lambda r. x) \in \mathcal{R}}$$

precomposition:

$$\frac{\alpha \in \mathcal{R} \quad \varphi \in \mathbf{Meas}(\mathbb{R}, \mathbb{R})}{(\alpha \circ \varphi) \in \mathcal{R}}$$

recombination:

$$\frac{I \subseteq \mathbb{N} \quad E : I \rightarrow \mathcal{B}_{\mathbb{R}} \quad \mathbb{R} = \bigcup_{i \in I} E_i \quad \alpha : I \rightarrow \mathcal{R}}{[E_i. \alpha_i]_{i \in I} := (\lambda r \in E_i. \alpha_i r) \in \mathcal{R}}$$



¹ $\mu\epsilon\tau\alpha$ ('meta', across) and $\varphi\epsilon\rho\omega$ ('phero', to carry).

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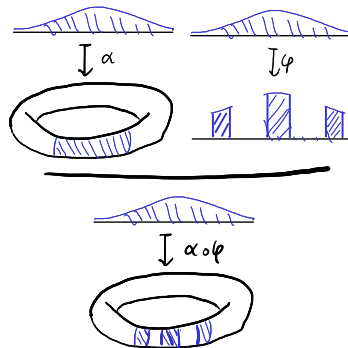
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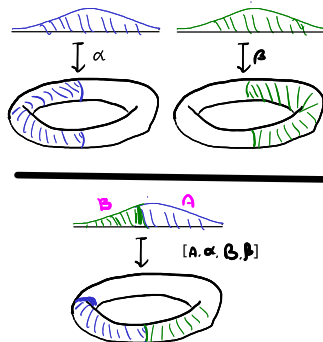
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Quasi Borel space (qbs) X

set of points $\underline{X}$ equipped with a metaphorology $\mathcal{R}_X$ over it

recombination:

$$\frac{I \subseteq \mathbb{N} \quad E : I \rightarrow \mathcal{B}_{\mathbb{R}} \quad \mathbb{R} = \bigsqcup_{i \in I} E_i \quad \alpha : I \rightarrow \mathcal{R}}{[E_i. \alpha_i]_{i \in I} := (\lambda r \in E_i. \alpha_i r) \in \mathcal{R}}$$

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Γ -fibred metaphorology $\mathcal{R}^-$ over $\gamma : \underline{\Gamma} \vdash \underline{X}_\gamma$

$\mathcal{R}_\Gamma$ -indexed family of subsets $\nu : \mathcal{R}_\Gamma \vdash \mathcal{R}^\nu \subseteq (r : \mathbb{R}) \Rightarrow \underline{X}_{\nu r}$ closed under:

fibred constants:

$$\frac{\gamma \in \underline{\Gamma} \quad a \in \underline{X}_\gamma}{\underline{a} := (\lambda r. a) \in \mathcal{R}^\gamma}$$

fibred precomposition:

$$\frac{\alpha \in \mathcal{R}^\nu \quad \varphi \in \mathbf{Qbs}(\mathbb{R}, \mathbb{R})}{(\alpha \circ \varphi) \in \mathcal{R}^{\nu \circ \varphi}}$$

fibred recombination:

$$\frac{I \subseteq \mathbb{N} \quad E : I \rightarrow \mathcal{B}_\mathbb{R} \quad \mathbb{R} = \bigsqcup_{i \in I} E_i \quad \alpha : (i : I) \rightarrow \mathcal{R}^{\nu_i}}{[E_i. \alpha_i]_{i \in I} \in \mathcal{R}^{[E_i. \nu_i]_{i \in I}}}$$

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Γ -fibred metaphorology $\mathcal{R}^-$ over $\gamma : \underline{\Gamma} \vdash \underline{X}_\gamma$

$\mathcal{R}_\Gamma$ -indexed family of subsets $\mathbf{v} : \mathcal{R}_\Gamma \vdash \mathcal{R}^{\mathbf{v}} \subseteq (r : \mathbb{R}) \Rightarrow \underline{X}_{\mathbf{v}r}$ closed under:

fibred constants: $\frac{\gamma \in \underline{\Gamma} \quad a \in \underline{X}_\gamma}{\underline{a} := (\lambda r. a) \in \mathcal{R}^\gamma}$	fibred precomposition: $\frac{\alpha \in \mathcal{R}^{\mathbf{v}} \quad \varphi \in \mathbf{Qbs}(\mathbb{R}, \mathbb{R})}{(\alpha \circ \varphi) \in \mathcal{R}^{\mathbf{v} \circ \varphi}}$
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fibred recombination:

$$\frac{I \subseteq \mathbb{N} \quad E : I \rightarrow \mathcal{B}_\mathbb{R} \quad \mathbb{R} = \bigsqcup_{i \in I} E_i \quad \alpha : (i : I) \rightarrow \mathcal{R}^{\mathbf{v}_i}}{[E_i.\alpha_i]_{i \in I} \in \mathcal{R}^{[E_i.\mathbf{v}_i]_{i \in I}}}$$

Γ -fibred metaphorology $\mathcal{R}^-$ over $\gamma : \underline{\Gamma} \vdash \underline{X}_\gamma$

$\mathcal{R}_\Gamma$ -indexed family of subsets $\nu : \mathcal{R}_\Gamma \vdash \mathcal{R}^\nu \subseteq (r : \mathbb{R}) \Rightarrow \underline{X}_{\nu r}$ closed under:

$$\begin{array}{ll} \text{fibred constants:} & \text{fibred precomposition:} \\ \frac{\gamma \in \underline{\Gamma} \quad a \in \underline{X}_\gamma}{\underline{a} := (\lambda r. a) \in \mathcal{R}^\gamma} & \frac{\alpha \in \mathcal{R}^\nu \quad \varphi \in \mathbf{Qbs}(\mathbb{R}, \mathbb{R})}{(\alpha \circ \varphi) \in \mathcal{R}^{\nu \circ \varphi}} \end{array}$$

fibred recombination:

$$\frac{I \subseteq \mathbb{N} \quad E : I \rightarrow \mathcal{B}_\mathbb{R} \quad \mathbb{R} = \bigsqcup_{i \in I} E_i \quad \alpha : (i : I) \rightarrow \mathcal{R}^{\nu_i}}{[E_i. \alpha_i]_{i \in I} \in \mathcal{R}^{[E_i. \nu_i]_{i \in I}}}$$

Quasi Borel family (qbf) $\Gamma \vdash X$

a family $\gamma : \underline{\Gamma} \vdash \underline{X}_\gamma$ equipped with a Γ -fibred metaphorology $\mathcal{R}_X$ over $\underline{X}$.

Quasi Borel families

Examples

► $\frac{\Gamma, X \text{ qbses}}{\Gamma \vdash X \text{ qbf}} \quad \gamma : \underline{\Gamma} \vdash \underline{X}_\gamma := X \quad v : \mathcal{R}_\Gamma \vdash \mathcal{R}_X^v := \mathcal{R}_X$

► Intervals are qbfs:

$$a, b : [-\infty, \infty] \vdash [a, b], [a, b), (a, b], (a, b)$$

with $\mathcal{R}^{a \mapsto \alpha, b \mapsto \beta}$ those measurable functions $\varphi \in \mathbf{Qbs}(\mathbb{R}, [0, \infty])$ such that, for all $r \in \mathbb{R}$, φr is in the interval delimited by αr and βr .

► For measurable map $\begin{smallmatrix} X \\ d \downarrow \\ \Gamma \end{smallmatrix}$, the **preimage** qbf:

$$\gamma : \Gamma \vdash d^{-1}[\gamma] \quad \mathcal{R}_{d^{-1}[-]}^v := \left(\begin{smallmatrix} \mathcal{R}_X \\ d \circ \downarrow \\ \mathcal{R}_\Gamma \end{smallmatrix} \right)^{-1} [v] \subseteq \mathcal{R}_X$$

Quasi-measurable maps

Qbf map $(\theta \vdash f) : (\gamma : \Gamma \vdash X_\gamma) \rightarrow (\delta : \Delta \vdash Y_\delta)$

a family map $(\theta \vdash f) : (\gamma : \underline{\Gamma} \vdash \underline{X}_\gamma) \rightarrow (\delta : \underline{\Delta} \vdash \underline{Y}_\delta)$, i.e.:

$$\theta : \underline{\Gamma} \rightarrow \underline{\Delta} \quad \gamma : \Gamma \vdash f_\gamma : \underline{X}_\gamma \rightarrow Y_{\theta\gamma}$$

such that:

- ▶ $\theta : \Gamma \rightarrow \Delta$ in **Qbs**, i.e., preserves random elements:

$$v \in \mathcal{R}_\Gamma \implies (\theta \circ v) \in \mathcal{R}_\Delta$$

- ▶ f preserves fibred random elements:

$$\mathcal{R}_X^v \ni \alpha \implies (f \overset{\theta}{\circ} \alpha) := (\lambda r. f_{\theta r}(\alpha r)) \in \mathcal{R}_Y^{\theta \circ v}$$

Theorem

The category of qbfs and their maps, $\mathbf{Qbf}$, equipped with the functor:

$$\text{Index} : (\Gamma \vdash X_-) \mapsto \Gamma : \mathbf{Qbf} \rightarrow \mathbf{Qbs}$$

is a split fibred category.

It supports:

- ▶ *split Π -types;*
- ▶ *Σ -types;*
- ▶ *split extensional propositional equality; and*
- ▶ *split subspaces.*

*The **Grothendieck construction** and the **preimage** form a fibred adjoint equivalence between Index and the codomain fibration $\mathbf{Qbs}^{\rightarrow} \rightarrow \mathbf{Qbs}$.*

Talk structure

- ▶ Motivation
- ▶ quasi-Borel spaces & families
- ▶ **fibred probability monads**
- ▶ dependently-typed probability: conditional expectation

Lebesgue integration of random elements

injective measurable map (monomorphism in **Qbs**):

$$\int d[-] := \left(\lambda f. \lambda g. \int_{\text{Dom}(f)} dx (g \circ f) \right) : \underbrace{(\mathbb{R} \rightarrow X^\perp)}_{=: \mathbf{R}X} \rightarrow \underbrace{(X \rightarrow \mathbb{W}) \rightarrow \mathbb{W}}_{=: \mathbf{K}X}$$

NB:

- ▶ measurable **partial functions** $f : X \multimap Y$ are measurable functions
 $f : X \multimap (Y \amalg \{\perp\}) =: Y^\perp$
- ▶ space of **weights**: $\mathbb{W} := [0, \infty]$

Lebesgue integration of random elements

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Distribution monad $\mathbf{D}X$

Strong epi-mono factorisation of $\int d[-]$:

$$\int d[-] : \mathbf{R}X \xrightarrow{[-]} \mathbf{D}X \xrightarrow{\int d} \mathbf{K}X \quad \text{concretely:} \quad \begin{aligned} \mathbf{D}X &:= \{[\alpha] \mid \alpha : \mathbb{R} \multimap X\} \\ \mathcal{R}_{\mathbf{D}X} &:= \left\{ \alpha : \mathbb{R} \rightarrow \mathbf{D}X \mid \begin{array}{l} \exists \beta \in \mathcal{R}_{\mathbf{R}X}. \\ \alpha = \lambda r. [\beta r] \end{array} \right\} \end{aligned}$$

Lebesgue integration of fibred random elements

injective measurable map (monomorphism in **Qbs**):

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Fibred distribution monad $\mathbf{D}X$

Fibred strong epi-mono factorisation of $\int d[-]$:

$$\Gamma \vdash \int d[-] : \mathbf{R}X \xrightarrow{[-]} \mathbf{D}X \xrightarrow{\int d} \mathbf{K}X$$

concretely:

$$\gamma : \underline{\Gamma} \vdash \underline{\mathbf{D}X}_\gamma := \{[\alpha] \mid \alpha : \mathbb{R} \multimap X_\gamma\} \quad \mathbf{v} : \mathcal{R}_\Gamma \vdash \mathcal{R}_{\mathbf{D}X}^{\mathbf{v}} := \left\{ \alpha : \mathbb{R} \rightarrow \mathbf{D}X \mid \begin{array}{l} \exists \beta \in \mathcal{R}_{\mathbf{R}X}^{\mathbf{v}}. \\ \alpha = \lambda r. [\beta r] \end{array} \right\}$$

Synthetic measure theory

Theorem

$\mathbf{D}$ and $\mathbf{P}X := \{\mu : \mathbf{D}X \mid 1 = \mu X := \int d\mu 1\}$ are commutative split monads on $\mathbf{Index}$.
The fibres of $\mathbf{D}$ satisfy Kock's [12] synthetic measure theory axioms—there are canonical isomorphisms:

$$\Gamma \vdash \mathbf{D} \left(\coprod_{i \in I} A^i \right) \xrightarrow{\cong} \prod_{i \in I} \mathbf{D}(A^i) \quad (I \text{ countable})$$

Consequences

Derived notions of, e.g.:

semi-ring of scalars: $\mathbb{W} = [0, \infty]$; expectation; disintegration; almost-sure equality;
density / absolute continuity / Radon-Nikodym derivatives; etc. [Kock'12]

compatible with measure-theoretic concepts [Ścibior et al. 18]

Talk structure

- ▶ Motivation
- ▶ quasi-Borel spaces & families
- ▶ fibred probability monads
- ▶ dependently-typed probability: conditional expectation

Dependently-typed probability

Use case

Study spaces of integrable random variables—**Lebesgue** space quasi-Borel **family**:

$$\frac{\Gamma \vdash \mu : \mathbf{P}\Omega \quad \Gamma \vdash p : [1, \infty)}{\Gamma \vdash \mathcal{L}^p(\Omega, \mu) := \left\{ f : \Omega \rightarrow \mathbb{R} \mid f \in \text{Dom} \left(\int \mathrm{d}\mu \right) \wedge \int \mathrm{d}\mu |f| < \infty \right\}}$$

Different from the quotiented **Banach** spaces:

$$\mathbf{L}^{p_\gamma}(\Omega_\gamma, \mu_\gamma) := \mathcal{L}^{p_\gamma}(\Omega_\gamma, \mu_\gamma) / (\mu_\gamma\text{-a.s. equality})$$

Conditional expectation

Central higher-order concept to modern probability:

$$\begin{array}{c}
 \Gamma \vdash \mu : \mathbf{D}\Omega \quad \Gamma \vdash H : \Omega \rightarrow \Theta \quad \Gamma \vdash f : \mathcal{L}^1(\Omega, \mu) \quad \Gamma \vdash g : \mathcal{L}^1(\Theta, \mu_H) \\
 \hline
 \Gamma \vdash (g = \mathbb{E}_\mu [f|H]) := \quad \forall w : \Theta \rightarrow \mathbb{W}. \\
 \int \mu_H(d\theta) g(\theta) \cdot w(\theta) = \int \mu(d\omega) f(\omega) \cdot w(H \omega) : \mathbf{Prop}
 \end{array}$$

Theorem

Let $\Gamma \vdash \Omega$ be any qbf and $\vdash \Theta$ a standard Borel space. We have a measurable dependent function:

$$\Gamma \vdash \mathbb{E} : (\mu : \mathbf{P}\Omega) \rightarrow (H : \Omega \rightarrow \Theta) \rightarrow (f : \mathcal{L}^1(\Omega, \mu)) \rightarrow \{g : \mathcal{L}^1(\Theta, \mu_H) \mid g = \mathbb{E}_\mu [f|H]\}$$

What have we done?

Best possible situation:

- ▶ Semantic model for dependent types (Π , Σ , extensional identity, subspaces)
(split fibration)
- ▶ Semantic model for dependently-typed probability
(split distribution and probability monads)

Case study

- ▶ dependently-typed probability
(measurability of conditional expectation under regularity assumptions)

Dependently-typed accounts of:

- ▶ disintegration;
- ▶ martingale theory;
- ▶ stochastic differential equations
(NB: generalized conditional expectation may need further assumptions)
- ▶ universe constructions
(already have $\mathbf{Sbs} := \mathcal{B}_{\mathbb{R}}$; $\mathbf{Pol}$ and $\mathbf{CSepMet}$ via Effros space)
- ▶ implications to Bayesian inference implementations;
- ▶ implications to probabilistic programming semantics
(operational and denotational)